

Bassam Jameel

PhD, Ultrasound Systems & Sensor Engineering

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Personal Profile

Ultrasound systems researcher with 5+ years of experience spanning transducer characterization, high-frequency optical-ultrasound sensor design, real-time RF signal processing, and acoustic field modelling in COMSOL Multiphysics. Hands-on across electronics design, sensor fabrication, and instrumentation, with a track record of 8 peer-reviewed publications, 1 invited talk, and 12 conference presentations.

Work and Research Experience

- **Postdoctoral researcher**

Fresnel Institute, Aix-Marseille University, CNRS

Marseille, France

01/04/2025 – present

- Numerical model of the optical-ultrasound sensors.
- Characterization of the mechanical properties and the film thickness using pulse echo system.
- Developing an optical-acoustic platform to measure overall sensitivity and frequency response.
- Image reconstruction in photoacoustic tomography.

- **Research engineer**

Institute of Experimental Physics, Slovak Academy of Sciences

Kosice, Slovakia

01/10/2022 – 01/04/2023

- Characterizing and synthesizing magnetic nanoparticles.
- Determining the magnetorheological effect in magnetic emulsions and fluids.

Teaching Experience

- **Assistant supervisor**

Acoustic Institute, Faculty of Physics, Adam Mickiewicz University

Poznan, Poland

15/02/2021 – 01/07/2021

- Molecular acoustic laboratory instructor.
- Assistant supervisor of master students' projects.

15/02/2022 – 01/07/2022

Education

- **01/10/2020 – 18/10/2024 Physics, PhD**

Acoustic Institute, Faculty of Physics, Adam Mickiewicz University, Poznan-Poland

- Develop ultrasound system to determine the Pickering droplet size and stability.
- Detecting the agglomeration rate in high-viscous-magnetic and non-magnetic suspensions.
- Optimization of ultrasound heating in tissue-mimicking phantoms using Pickering droplet.
- Controlling the magnetic separation and heating in magnetic Pickering emulsions by ultrasound.

- **01/10/2018 – 21/07/2020 Physics, MSc**

Faculty of Physics, Adam Mickiewicz University, Poznan-Poland

Program: Physics of advanced materials for energy processing

Focus: Nanomaterial characterization and microwave engineering

- Simulation and fabrication of low-pass microwave filters based on strip transmission line technology.
- Optimization the cut-off frequency of microwave filter with different terminal orders for strip lines.

- **01/10/2011 – 15/06/2015 Physics, BSc**

Faculty of Science, Department of Physics, Al-Mustansiriya University, Baghdad-Iraq

Focus: General Physics

Technical Skills

Ultrasound signal processing, pulse-echo ultrasonic testing, piezoelectric transducer characterization, phased-array beamforming, acoustic field modelling (COMSOL and k-Wave) and ultrasound sensor simulation, sensor metrology and characterization, data analysis, digital image processing, 3D structure design, 3D printing, model implementation, and microwave filter design and simulation.

Software Skills

MATLAB, Python (NumPy, Pandas, SciPy, Scikit-learn), AutoCAD, Blender, Wolfram Mathematica, COMSOL Multiphysics, CST Microwave Studio, and ImageJ.

Laboratory Skills

Ultrasound system, photoacoustic system, AC magnetic field, rotating magnetic field, electronic circuit design, magnetic fluid fabrication, Pickering emulsion fabrication, rheometer, photolithography, thin film fabrication, and optical microscopy.

Soft Skills

Self-learner, problem-solving, analytical thinking, time management, creativity, and adaptability.

Scientific Publications ([ResearchGate](#))

1. **Jameel, B.**, Harkavyi, Y., Bielas, R., and Józefczak, A., "Optimization of ultrasound heating with Pickering droplets using core-shell scattering theory", **Ultrasonic sonochemistry**, (2024): 106965, doi: [10.1016/j.ultsonch.2024.106965](https://doi.org/10.1016/j.ultsonch.2024.106965)
2. Ratajczak, F., **Jameel, B.**, Bielas, R., and Józefczak, A., "Ultrasound control of Pickering- emulsion based capsule preparation", **Sensors** 24.17 (2024): 5710, doi: [10.3390/s24175710](https://doi.org/10.3390/s24175710)
3. **Jameel, B.**, Paulovičová, K., Tóthová, J., Rajňák, M., Molčan, M., Bielas, R. and Józefczak, A. "Magnetorheology characterization of oil-in-oil magnetic Pickering emulsions", **Journal of Magnetism and Magnetic Materials** (2023): 171433, doi: [10.1016/j.jmmm.2023.171433](https://doi.org/10.1016/j.jmmm.2023.171433)
4. **Jameel, B.**, Bielas, R. and Józefczak, A. "Ultrasound measurements of particle shells in magnetic Pickering emulsions", **Measurement** (2023): 113409, doi: [10.1016/j.measurement.2023.113409](https://doi.org/10.1016/j.measurement.2023.113409)
5. **Jameel, B.**, Hornowski, T., Bielas, R. and Józefczak, A. "Ultrasound study of magnetic and non-magnetic nanoparticles agglomeration in high viscous media", **Materials**, 15(10), 3450. 2022, doi: [10.3390/ma15103450](https://doi.org/10.3390/ma15103450)
6. **Jameel, B.**, Hornowski, T., Bielas, R. and Józefczak, A. "Propagation of ultrasonic wave in magnetic Pickering emulsion under DC magnetic field", **Journal of Magnetism and Magnetic Materials** 542(2022): 168590, doi: [10.1016/j.jmmm.2021.168590](https://doi.org/10.1016/j.jmmm.2021.168590)
7. Bielas, R., **Jameel, B.**, Skumiel, A., Timko, M., Kopčanský, P. and Józefczak, A. "Magnetic Pickering emulsions heated in a rotating magnetic field", **Journal of Magnetism and Magnetic Materials** 563 (2022): 169946, doi: [10.1016/j.jmmm.2022.169946](https://doi.org/10.1016/j.jmmm.2022.169946)
8. Bielas, R., **Jameel, B.**, and Józefczak, A., "Monitoring of Pickering emulsion stability during magnetic heating using ultrasound measurements", **Measurement** 178 (2021): 109431, doi: [10.1016/j.measurement.2021.109431](https://doi.org/10.1016/j.measurement.2021.109431)

Conference Presentations

Total number of presentations:12

Invited talks: 1, Oral presentations: 8, Poster presentations: 4. The following selected presentations:

1. Image Science Gordon Research Seminar (GRS) and conference (GRC), 25.04.2026 – 26.04.2026

Oral presentation (GRS): Sensitivity Optimization of Optical Ultrasound Sensors for Photoacoustic Imaging

- Poster presentation** (GRC): Sensitivity Optimization of Optical Ultrasound Sensors for Photoacoustic Imaging
2. 14th International Conference on the Scientific and Clinical Applications of Magnetic Carriers (Barcelona, Spain), 17.06.2024 – 21.06.2024
Poster presentation: Pickering droplets stabilized by magnetic nanoparticles for optimizing ultrasound heating in tissue-like medium
3. International Conference on Organic and Nanohybrid Functional Materials (Poland, Gdansk), 10.06.2024 – 14.06.2024
Oral presentation: Magnetic and non-magnetic Pickering droplets for ultrasound hyperthermia optimization
4. 69th Open Seminar on Acoustics (Poland, Karpaz), 25.09.2023 – 29.09.2023
Invited talks: Magnetic Pickering emulsions characterization by ultrasound attenuation spectroscopy.
5. International Conference on Magnetic Fluid (Spain, Granada), 12.06.2023 – 16.06.2023
Poster presentation: Magnetorheological properties of oil-in-oil magnetic Pickering emulsion.
6. 51st Winter School on Wave and Quantum Acoustics (Poland, Szczyrk), 27.02.2023 – 03.03.2023
Oral presentation: Ultrasound study in magnetic Pickering emulsion.
7. 22nd International Workshop on Theoretical Physics/ Small Triangle Meeting (Slovakia, Medzilaborce), 24.10.2022 – 27.10.2022
Oral presentation: Ultrasound study of high-viscous oil magnetic and non-magnetic suspensions.
8. NanoTech Poland 2021 (Poland, Poznan), 09.06.2021 – 11.06.2021
Oral presentation: Ultrasound spectroscopy for studying Pickering emulsion stabilized by magnetic nanoparticles

Training and workshop

- [International Summer School on Advanced Ultrasound Imaging](#), Technical University of Denmark, 22.05.2024 – 31.05.2024
- [Biomedical Imaging School 2025](#), Cargese, Corsica, 14.04.2025–18.04.2025
- Acoustics Modeling in COMSOL Multiphysics, COMSOL, 27.05.2025–31.05.2025
- [Training Workshop - AI for Imaging 2025](#), Marseille, France, 15.12.2025–18.12.2025

Scientific blogs

- Delay and Sum Beamforming of Ultrasound Imaging Using Simulated Data
Notebook link: <https://brick-flyingfish-037.notion.site/Delay-and-Sum-Beamforming-in-Ultrasound-Imaging-164c295e8a50802c8593ed3765cce771>
- Delay and Sum Beamforming of Ultrasound Imaging Using Experimental Data
Notebook link: <https://www.notion.so/Delay-and-Sum-Beamforming-of-Ultrasound-Imaging-Using-Experimental-Data-170c295e8a50805ab118f3c274d59b0a>
- List of the projects of integration system with GitHub link for each one for further explanation
Page Link: <https://bassam266.github.io/bassam-jameel/projects.html>

Awards and recognitions

- First-degree Rector Award for scientific achievements in 2022, 2024, and 2025
Adam Mickiewicz University

Peer Reviewer

- Acta Acustica (x2)
- Journal of Magnetism and Magnetic Material (x1)

References

Dr. Thomas Chaigne — Fresnel Institute, Aix-Marseille University. thomas.chaigne@fresnel.fr

Prof. Arkadiusz Józefczak — Faculty of Physics, Adam Mickiewicz University. arkadiusz.jozefczak@amu.edu.pl

Prof. Milan Timko — Institute of Experimental Physics, Slovak Academy of Sciences. timko@saske.sk